

AI/ML & LLM Glossary (Cheat Sheet)

Artificial Intelligence (AI): Simulation of human intelligence by machines (e.g., Siri, self-driving cars).

Machine Learning (ML): AI subset where systems learn patterns from data (e.g., spam filters).

Deep Learning: ML using neural networks with many layers (e.g., image recognition).

Supervised Learning: Model trained on labeled data (e.g., predicting house prices).

Unsupervised Learning: Model finds hidden patterns in unlabeled data (e.g., clustering customers).

Reinforcement Learning: Learning via trial & error using rewards (e.g., game-playing bots).

Neural Network: A system inspired by the brain with interconnected nodes (e.g., handwriting recognition).

NLP (Natural Language Processing): AI for understanding human language (e.g., Google Translate).

Computer Vision: AI for image & video understanding (e.g., facial recognition).

Overfitting: When a model memorizes data instead of generalizing (bad performance on new data).

Bias-Variance Tradeoff: Balance between underfitting and overfitting.

Big Data: Very large data sets needing special tools (e.g., Hadoop, Spark).

Data Preprocessing: Cleaning and preparing raw data for ML.

Feature Engineering: Creating input features to improve ML performance.

Model Training: Teaching an algorithm with data.

Model Inference: Using a trained model to make predictions.

LLM (Large Language Model): Trained on massive text to generate language (e.g., ChatGPT, Gemini).

Transformer: Neural network architecture powering LLMs (e.g., BERT, GPT).

Attention Mechanism: Helps models focus on important words in context.

Fine-tuning: Adjusting pre-trained models for specific tasks (e.g., legal AI).

Prompt Engineering: Designing prompts to guide LLM outputs.

Hallucination (AI): When AI outputs false but confident answers.

Embedding: Converting text/images into numeric vectors for meaning comparison.

Token: The smallest text unit processed by an LLM (e.g., 'ChatGPT' → 'Chat', 'GPT').

Context Window: Max tokens an LLM can remember in one session.

Zero-shot Learning: Performing tasks without task-specific training.

Few-shot Learning: Using a few examples in prompts to guide the model.

RLHF: Reinforcement Learning with Human Feedback (aligns AI with human values).

Generative AI: AI that creates new content (text, images, video).

Diffusion Models: Image generation models denoising random noise (e.g., Stable Diffusion).

Multimodal AI: AI that processes text, image, audio, and video together (e.g., GPT-4V).